MILITARY SPECIFICATION

RADIAC SET AN/GDS-1

This specification is mandatory
for use by USAF

1. SCOPE. This specification covers the requirements for a remote area fallout monitor designated as the AN/GDS-1 hereafter referred to as the Radiac Set. This Radiac Set is designed to monitor (remotely) gamma radiation from nuclear event fallout. The Radiac Set will be used at airforce bases and sites for monitoring the radiation environments surrounding fallout shelters and command posts.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

2.1.1 SPECIFICATIONS

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-I-8500	Interchangeability and Replaceability of Component parts for aircraft and missiles
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

2.1.2 STANDARDS

Military

MIL-STD-16	Electric and Electronic Reference Designations
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

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MIL-STD-810	Environmental Test Methods
MIL-STD-781A	Reliability Tests Exponential Distribution
MIL-STD-831	Test Report, Preparation of

2.1.3 DRAWINGS

7014000 (USAF)	Radiac Set, AN/GDS-1

(Copies of the above documents which are required by contractors in connection with specific procurement functions should be obtained from the Administrative Contracting Officer or as directed by the Procuring Activity.)

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provision for preproduction testing.

3.2 DESIGN AND CONSTRUCTION. The test set shall be designed and constructed in conformance to Drawing No. 7014000 and subsidiary drawings.

3.2.1 INTERCHANGEABILITY. Parts functionally and dimensionally interchangeable shall be identified by the same part number. When parts are not interchangeable, the criteria of MIL-D-1000 shall apply and new drawings shall be prepared as required. Physical interchangeability shall be in accordance with MIL-I-8500.

3.3 PERFORMANCE. Satisfactory performance of the Radiac Set shall consist of the capability to indicate the actual field strength to within $\pm$20% for a gamma dose rate of .1 R/hr. to 10,000 R/hr. under the environmental conditions specified in this specification.

3.3.1 RELIABILITY. The Radiac Set shall have a Mean Time Before Failure (MTBF) rating of 250 cycles of operation under service conditions.

3.4 ENVIRONMENTAL REQUIREMENTS. The Radiac Set shall be capable of satisfactory performance during or after, as applicable, subjection to the following environmental conditions:

a. High temperatures to 120°F.

b. Low temperatures to -40°F.

c. Temperature shock incident to service use.

d. Altitude to 20,000 feet.

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e. Humidity incident to storage or service use in a damp climate.

f. Fungus such as encountered in storage or use in tropical climates.

g. Shock incident to transportation, handling and service conditions.

h. Solar radiation incident to service use.

i. Rain incident to service use.

j. Vibration. The Radiac Set shall be capable of satisfactory operation after vibration, as specified in vibration tests of this specification.

3.5 IDENTIFICATION OF PRODUCT. The Radiac Set, parts and assemblies shall be marked for identification in accordance with MIL-STD-130.

3.6 REFERENCE DESIGNATIONS. Reference designations used for marking within the unit shall be in accordance with MIL-STD-16. Markings shall not adversely affect the leakage path between conductors, or any other factor of performance. Reference symbols shall be employed to identify each part for its particular circuit application. The same reference symbol shall be used for identifying a particular part on all types of data, including drawings, diagrams, and spare parts lists prepared for the equipment. When practicable, markings shall be placed on the supporting structure adjacent to each replaceable part in such a position as to be readily reviewed.

3.7 WORKMANSHIP. The Radiac Set shall be fabricated and finished in a manner such that the criteria of appearance, fit and adherence to specified tolerances shall be observed. These criteria shall apply to all nonfunctioning parts as well as to functioning parts. Particular attention shall be given to making of parts and assemblies, plating, machine screw assemblage, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. Inspection and testing of the Radiac Set shall be classified as follows:

a. Preproduction tests.

b. Acceptance tests.

4.2 PREPRODUCTION TESTING

4.2.1 PREPRODUCTION TEST SAMPLES. The preproduction test units shall be representative of the production equipment. They shall be tested by a laboratory designated by the procuring activity, or when so stated in the contract, at the contractor's facility under the supervision of the procuring activity.

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4.2.1.1 PREPRODUCTION REPORT. In the event that preproduction testing is performed at the contractor's facility, the contractor shall, at the completion of preproduction tests, prepare a preproduction test report. The preproduction test report shall be prepared in accordance with MIL-STD-831 and three complete copies shall be furnished the procuring activity.

4.2.2 PREPRODUCTION TESTS. Preproduction tests shall consist of all tests described in "4.5 TEST METHODS" of this specification.

4.3 ACCEPTANCE TESTS. Acceptance tests shall consist of the following:

a. Individual tests.

b. Sampling tests.

4.3.1 INDIVIDUAL TESTS. Each Radiac Set shall be subjected to the following individual tests:

a. Examination of product.

b. Conformance to drawings.

c. Functional operational test.

4.3.2 SAMPLING TESTS. Sampling tests shall consist of the following and shall be performed in the order shown:

a. Low Temperature Test.

b. High Temperature Test.

c. Vibration Test.

4.3.2.1 SAMPLING PLAN. Radiac Sets to be subjected to the sampling tests shall be selected at random from production items as follows:

a. First 100 produced, 1 sample from each lot of 10.

b. Each additional 1000 produced, 1 sample from each lot of 100.

4.3.2.2 REJECTION AND RETEST. When a sample item fails to satisfy the specified tests, no item still on hand or later produced shall be accepted until the extent and cause of failure is determined and resolved.

4.3.2.3 INDIVIDUAL TESTS MAY CONTINUE. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or produced later shall not be made until it is determined that the items meet all requirements of this specification.

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4.3.3 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a test failure could indicate that defects exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 TEST CONDITIONS. Unless otherwise specified, tests shall be performed at room atmospheric conditions, and the unit shall be at room temperature when each test is started.

4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT. Each Radiac Set shall be examined visually to determine compliance with all requirements of this specification for which tests are not specified. Failure to meet these requirements shall be cause for rejection.

4.5.2 CONFORMANCE TO DRAWINGS. Each Radiac Set shall be inspected to determine conformance to the engineering drawings, subsidiary drawings and all specifications and standards referenced therein. Failure to conform to these criteria shall be cause for rejection.

4.5.3 FUNCTIONAL TESTS. Functional tests shall be performed on each Radiac Set in accordance with Table I to ensure compliance with this specification.

4.5.4 ENVIRONMENTAL TESTS. Pretest data shall be taken for all environmental tests and shall consist of a calibration with a Co^{60} source. During or after subjection to the environmental tests, as specified in the test, the Radiac Set shall be subjected to the functional test. Failure to satisfy the requirements of the functional test shall be considered as failure to pass the environmental test and cause for rejection.

4.5.4.1 HIGH TEMPERATURE TEST. The Radiac Set shall be subjected to the high temperature test of Method 501, Procedure I, MIL-STD-810. The functional test shall be performed after 24 hours exposure and after 48 hours of exposure. While at the specific temperature, a reference point (between 5 R/hr. and 10 R/hr.) on the calibration curve shall be checked using an appropriate radiation source.

4.5.4.2 LOW TEMPERATURE. The low temperature test shall be as described in Method 502, Procedure I, MIL-STD-810. At the conclusion of this test, the functional test shall be performed. While the Radiac Set is at the specified low temperature, a reference point (between 5 R/hr. and 10 R/hr.) on the calibration curve shall be checked using an appropriate radiation source.

4.5.4.3 TEMPERATURE SHOCK TEST. The temperature shock test shall be as described in Method 503, Procedure I, MIL-STD-810. At the conclusion of the shock test and after the instrument has been returned to room temperature, the functional test shall be performed.

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TABLE I

FUNCTIONAL TEST

1. PRETEST CONDITIONS. The Radiac Set shall be calibrated prior to performing any environmental testing.

2. FUNCTIONAL CHECK. A functional check of the Radiac Set operation will be performed as shown in the following table.

Controls	Position	Gamma Radiation Field	Meter Reading	Comments
Range Switch	1	5 R/hr. to 10 R/hr.	5 R/hr. to 10 R/hr.	Meter Reading shall read within ±20% of a true logarithmic reading.
Power	AC			

3. FUNCTIONAL TEST. The functional test shall consist of an accuracy check over the 5 decades of instrument range. Allow a thirty minute warm-up if the Radiac Set has not been on.

Controls	Position	Gamma Radiation Field	Meter Reading	Comments
Range Switch	.1	.1 R/hr.	.1 R/hr.	Power Switch on AC
	.1	.5 R/hr.	.5 R/hr.	
	.1	1 R/hr.	1 R/hr.	The indicated meter reading shall be within +20% of a true logarithmic reading.
	1	1 R/hr.	1 R/hr.	
	1	5 R/hr.	5 R/hr.	
	1	10 R/hr.	10 R/hr.	
	10	10 R/hr.	10 R/hr.	
	10	50 R/hr.	50 R/hr.	
	10	100 R/hr.	100 R/hr.	
	100	100 R/hr.	100 R/hr.	
	100	500 R/hr.	500 R/hr.	
	100	1,000 R/hr.	1,000 R/hr.	
	1,000	1,000 R/hr.	1,000 R/hr.	
	1,000	5,000 R/hr.	5,000 R/hr.	
	1,000	10,000 R/hr.	10,000 R/hr.	
Power measurement shall be 32 ma ± ma at 115 V AC.				

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4.5.4.4 ALTITUDE TEST. The altitude test shall be as described in Method 500, Procedure I, MIL-STD-810. Maximum altitude shall be 20,000 feet.

When the Radiac Set is at the specified altitude, a reference point (between 5 R/hr. and 10 R/hr.) on the calibration curve shall be checked using an appropriate radiation source. At the conclusion of the altitude test, a functional test shall be performed.

4.5.4.5 HUMIDITY TEST. The humidity test shall be performed in accordance with Method 507, Procedure IV, MIL-STD-810. The high temperature shall be 120°F. At the points specified in MIL-STD-810, a reference point (between 5 R/hr. and 10 R/hr.) on the calibration curve shall be checked using an appropriate radiation source. A functional test shall be performed at the conclusion of the humidity test.

4.5.4.6 FUNGUS RESISTANCE TEST. The fungus resistance test shall be as described in Method 508, Procedure I, MIL-STD-810. Equipment operation shall consist of the functional test after completion of test.

4.5.4.7 SOLAR RADIATION TEST. The solar radiation test shall be performed in accordance with Method 505, Procedure I, MIL-STD-810. A functional test shall be performed after completion of this test. A reference point (between 5 R/hr. and 10 R/hr.) shall be checked using an appropriate radiation source during this test.

4.5.4.8 RAIN TEST. The rain test shall be performed in accordance with Method 506, Procedure I, MIL-STD-810. A functional test shall be performed at conclusion of this test.

4.5.4.9 VIBRATION TEST. Vibration testing of the Radiac Set shall be performed in accordance with Method 514, Procedure XI, MIL-STD-810. A functional test shall be performed at conclusion of this test.

4.5.4.10 RELIABILITY TEST. The Radiac Set shall be subjected to a reliability test to determine Mean Time Before Failure and expected service life. The reliability test shall consist of performance of a functional check repeated 350 times. Each performance of the functional check shall be considered one cycle. The test procedure and environmental conditions as specified in the following paragraphs shall be used for the reliability test.

4.5.4.10.1 TEMPERATURE CYCLING. TEMPERATURE cycling shall be two hours to stabilize at -40°F followed by two hours to stabilize at +25°F followed by two hours to stabilize at +120°F, plus two hours of functional check time after each stabilization period. The functional check shall consist of checking a reference point (between 5 R/hr. and 10 R/hr.) using an appropriate radiation source. Each four hours of temperature test time plus the functional check shall be considered one cycle.

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4.5.4.10.2 VIBRATION. The Radiac Set shall be vibrated per test level E of MIL-STD-781A.

Failure of any type prior to 175 cycles shall be cause for rejection. Failure after 175 cycles and before 350 cycles requiring adjustment by means other than nominal controls shall be cause for rejection.

5. PREPARATION FOR DELIVERY. Preparation of the Radiac Set for delivery shall be in accordance with MIL-E-17555.

5.1 MARKING. Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

RADIAC SET AN/GDS-1

6. NOTES

6.1 INTENDED USE. The Radiac Set, AN/GDS-1, is intended for use as a remote area fallout monitor to monitor gamma radiation from nuclear events. The Radiac Set will be used at Air Force bases and sites for monitoring the radiation environments surrounding fallout shelters and command posts.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be performed by the procuring activity.

c. Levels of preservation, packaging and packing.

6.2.1 PREPRODUCTION SAMPLES. It is expected that the contract or purchase order will specify that a minimum of ten units representative of the production Radiac Set will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of inspection for these tests.

6.3 PRODUCTION DELIVERY. No production Radiac Set should be delivered prior to full approval of the preproduction samples.

6.4 PREPRODUCTION TESTS. Preproduction tests are those tests performed prior to production on representative samples of the production equipment to demonstrate that the equipment meets the requirements of this specification.

a. Acceptance Tests. Acceptance tests are those tests accomplished on representative samples of equipment submitted for acceptance under a

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contract to determine conformance to the requirements of this specification, and to determine that the equipment is equivalent in performance and construction to the approved preproduction samples.

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Preparing Activity
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Project No. 6665-F300

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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